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EBU5304 A

Joint Programme Examinations 2018/19

EBU5304 Software Engineering

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number	
BUPT student number	
Class number	

NOT allowed: electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Ling Ma, Dr Gokop Goteng, Dr Matthew Huntbach, Dr Karen Shoop

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Filename: 1819_EBU5304_A No answer book required

Software Engineering EBU5304

Previous test on QMPLUS 1819A

Solutions & Analysis

By Puyuan Zheng, BUPTIS.

(写在前面的话：解答题给分点均根据解析者本人对题目的理解认为的采分点。与实际若有不同，请以实际判分为准，本解析写出的采分点仅供参考。本解析若有问题错误，欢迎批评指正。)

Question 1: (33 Marks)

a) Answer the following questions: (11 Marks)

i) Explain portability in software development. (2 Marks)

【Answer】

Number of target systems software can be operated on.

Percentage of target dependent statements for software.

【考点】Portability 含义

【解析】

本题直接问 software development 中 portability 的含义，我们可知，portability 就是可移植，也就是可以运行在不同操作系统以及依赖程度。

Number of target systems software can be operated on.

Percentage of target dependent statements for software.

(2 Marks)

ii) Suggest the most appropriate software process model that might be used as a basis for developing an interactive travel planning system that helps users plan journeys. Give your reasons. (3 Marks)

【Answer】

Evolutionary Development Model

Because Evolutionary Development Model have rapid feedback, which specification can be developed incrementally for user feedback.

【考点】Software process model

【解析】

本题考查根据具体软件情景判断软件的模型。

根据题目，我们可知该软件为互动旅行规划系统，重点词汇，互动。

涉及到互动，显然我们想到的是 Evolutionary development Model。

接着回答原因：

Because Evolutionary Development Model have **rapid feedback**, which specification can be **developed incrementally for user feedback**.

(3 Marks,回答错模型 0 分，只回答正确模型 1 分)

Evolutionary development

- Activities are interleaved
- Rapid feedback
- Refining through many versions, evolves over time
  - Completion of a comprehensive product is impossible.
  - Deliver core functions to meet competitive or business pressure.
  - Core requirements are well understood but not the detailed extension.

Evolutionary development – suitable projects

- Suitable for:
  - small or medium-size interactive systems
  - parts of large systems, e.g. the user interface
- For short-lifetime systems.
- Project with multiple features and therefore releases.

Examples:
Social networking, communication, phone apps

【*reference*】*ppt02 p17, p20*

【简单总结】Software Process Model

只说核心点！

Waterfall 必须按顺序进行，某一步没有进行完不会下一步。适用于稳定性较高，国防安全相关的都是这个软件模型。

Evolutionary Development, 适用于交互，中小型短期的。

RUP (Rational Unified Process)，属于普适化的模型，没有鲜明应用类型。

Agile model：灵活性，客户参与，快速交付和持续改进

迭代和增量的开发方法。

iii) Fill in the blanks.

To capture the requirements, we focus on the _____customers_____ view of the system, we must use the language of requirement specification to describe these results.

In analysis, the focus shifts to the _____developer_____ view of the system, we must use the language of a set of analysis models to describe these results.

(4 Marks)

【Answer】

customer requirement specification

developer a set of analysis models

【考点】Requirements 和 Analysis 流程

【解析】

根据题意我们需要进行填空，关注什么系统什么视角以及使用什么东西表达结果。就是视角+结果。

不难得知，捕捉需求是要从 customer 视角，requirement specification 作为结果描述；

分析是从 developer 的视角，a set of analysis models 作为结果描述。

答案依次是：customer requirement specification

developer a set of analysis models

(4 Marks, 1 for each blank)

User Stories

- In Agile process, user requirements are expressed as **user stories**.
  - helps shift the focus from writing about requirements to talking about them.
- These are written on cards, called **story cards**.
- All agile user stories include a written sentence or two and, more importantly, **a series of conversations** about the desired functionality.

Why "Analyse"

- Focus shifts to developer and system internals.
  - Refining requirements.
- Aim: precise understanding of requirements.
  - Process of structuring requirements:
    - Understand
    - Change
    - Reuse
    - Maintain

Requirement specification

- Defining what the system should do;
- Aim: a complete description of the problem and of the constraints imposed by/on the environment
- Description may contain:
  - Functions of the system
  - Future extensions
  - Amount of documentation required
  - Response time and performance
  - Acceptance criteria
- Result: Requirements specification

Analysis

- Aim: analyse requirements to create a conceptual model of the software system.
  - Data modelling
  - Functional modelling and information/control flow
  - Behavioural modelling and state
  - User interface modelling
- Result: a set of Analysis Models

【reference】ppt05 p2; ppt06 p6; ppt02 p3 p4

iv) What is an integration build plan? (2 Marks)

【Answer】

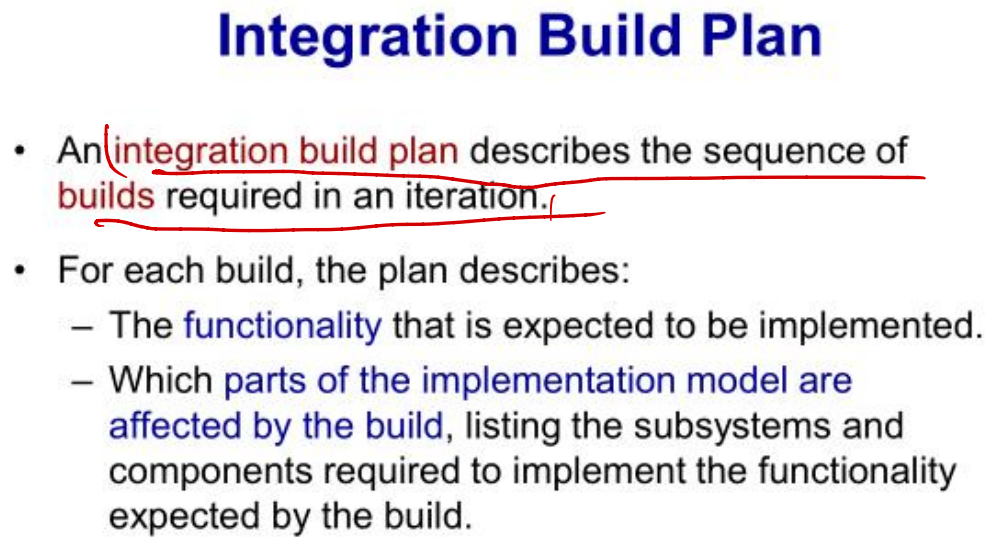
An integration build plan describes the sequence of builds required in an iteration.

【考点】Integration Build Plan

【解析】

本题问什么是 integration build plan，直接回答定义即可。

An integration build plan describes the sequence of builds required in an iteration. 在一轮迭代中需求的构建顺序



【reference】ppt08 p8

b) Answer the following questions about requirements: (14 Marks)

i) One of the problems of having a customer closely involved with a software development team is that they are influenced by the development team and lose sight of their real needs. One approach to avoid this problem is to use a different customer at each development stage. Discuss the advantages and disadvantages of this approach. (4 Marks)

【Answer】

Advantages:

Different customers have different view to the system, different perspective and different priorities.

Disadvantages:

Customer don’ t really know what is required for the software.

Customer may not have good understanding of the benefit of features and functions.

Maybe have difficulties in communication between different customers and development team.

【考点】Requirement

【解析】

本题给出获取需求时的问题——同一客户加入开发团队，他们的思想会被开发团队影响并且因此失去他们真实的想法。避免这种情况，我们采用不同阶段不同客户。分析优缺点。不同客户肯定有不同看待问题的视角和不同优先级；问题在于，客户能否真正理解开发者的意思，如何形成良好的沟通等。客户其实就是 stakeholder 的一类人，和 stakeholder 可以采用同样的答题模式。

Advantages:

Different customers have different view to the system, different perspective and different priorities.

Disadvantages:

Customer don’ t really know what is required for the software.

Customer may not have good understanding of the benefit of features and functions.

Maybe have difficulties in communication between different customers and development team.

(4 Marks)

Stakeholder

- Any person or organization who is affected by the system in some way and so who has a legitimate interest
- Stakeholders view the system differently
  - perspective
  - priorities

NOTE: A system may not always have an identifiable stakeholder who sets the requirements. Some software is developed according to an organisation's perception of market demand.

The problems

- Understanding the requirements is among the most difficult tasks that face a software engineer:
  - Does the customer know what is required?
  - Does the end-user have a good understanding of the features and functions that will provide benefit?
  - What they said = what they mean?
  - Good communication?
  - Even if the answers are all yes: requirements will change, and will continue to change throughout the project!

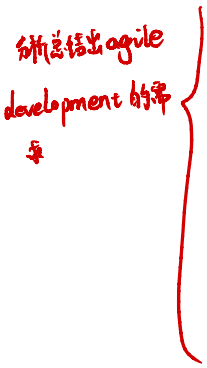
【*reference*】*ppt04 p5 p7*

ii) An Agile team is developing an automated ticketing system. The user story they are going to assess in the current iteration is "collecting pre-paid rail tickets". The user story is written as below:

An automated ticket machine can be used to collect pre-paid rail tickets. Users first press "Collect" on the touch screen, then they are prompted to insert the credit card they used to pay for the ticket. After that, users need to type in the unique Collection Reference Number they received. The machine checks the validity of the credit card and the Collection Reference Number. The ticket will then be dispensed.

Write three questions that the Agile team could discuss during the meeting in order to discover ambiguities or omissions in this story. (6 Marks)

【Answer】



Question 1: For the time a credit card bought for more than one ticket or not buy ticket, how to deal with it?

Question 2: As a user, how to receive the unique Collection Reference Number?

Question 3: How to check the validity for machine? What is the standard of valid or invalid?

【考点】Agile requirement

【解析】

本题根据情景开放性设计问题，agile team 需要讨论的，与这个 user story 相关的，是否有模糊或者省略的部分。

通过阅读这段 story 不妨分成如下几部分。

Users first press "Collect" on the touch screen, then they are prompted to insert the credit card they used to pay for the ticket.

After that, users need to type in the unique Collection Reference Number they received.

The machine checks the validity of the credit card and the Collection Reference Number.

The ticket will then be dispensed.

第一部分，如果一张 credit card 提前买了多张票或者没有买票的情况如何处理？省略细节。

第二部分，关于 number 用户如何获得没有提及，只说了需要输入 number，省略。

第三部分，机器如何检验 validity，模糊，没说具体。

综上分析，我们可以提出如下问题：

Question 1: For the time a credit card bought for more than one ticket or not buy ticket, how to deal with it?

Question 2: As a user, how to receive the unique Collection Reference Number?

Question 3: How to check the validity for machine? What is the standard of valid or invalid?

(6 Marks, 2 for each)

iii) Write two non-functional requirements for the above automated ticketing system, setting out its expected reliability and response time. (4 Marks)

【Answer】

The system should have a rate of availability of more than 99.999999%.

The system should have response time for less than 1 second.

【考点】Non-functional requirements

【解析】

本题要求根据上述自动售票系统设计两个非功能需求。

要求达到期待的可靠性和响应时间。

设计核心在于具体要多可靠？要多少响应时间？**要有具体指标！！！**

最好设计可靠性，出票准确率；最好设计响应时间，具体多少。根据生活常识设计即可。

出票准

The system should have a rate of availability of more than 99.999999%.

The system should have response time for less than 1 second.

(4 Marks, 2 for each).

响应快

【回顾】Non-functional requirement measures

Requirements measures

Property	Measure
Speed	Event response time Transactions/second Auto refresh time
Size	MB/GB
Ease of use	Training time Pages of instructions
Reliability	Rate of availbility Mean time to failure Probability of unavailability Rate of failure occurrence
Robustness	Time to restart after failure Percentage of events causing failure
Portability	Number of target systems Percentage of target dependent statements

【reference】ppt04 p20

c) Answer the following questions about using JUnit in testing: (8 Marks)

i) Explain how the use of automated tests and a testing framework such as JUnit simplify regression testing. (2 Marks)

【Answer】

Regression testing is an incremental development of test.

Using the testing framework like JUnit using a small cycle, each time when writing a specified code and in the form of unit test, The test verifies a functional unit of the code.

The process of development and testing can be in the same cycle, which is much simpler than the traditional way.

【考点】TDD, Regression testing

【解析】

本题问如何利用像 JUnit 这样的测试框架简化 regression testing。

首先我们从 regression testing 的本质出发，之后回答利用这种方式怎么简化。

Regression testing is an incremental development of test.

Using the testing framework like JUnit using a small cycle, each time when writing a specified code and in the form of unit test, The test verifies a functional unit of the code.

The process of development and testing can be in the same cycle, which is much simpler than the traditional way.

(2 Marks)

Regression Testing

- For integration testing.
- At each build stage, test cases will be created to test the functionality of the build.
- As this development is incremental, as new functionality is added to the build, so there are increasing numbers of test cases.
- These test cases will be used at each stage of iteration and they make up what are known as Regression Tests.
- The concept is that you continue to run all of the tests for each build to ensure that earlier functionality is not broken by updates to the build.

TDD cycle

- TDD: write tests prior to write the production code
- TDD is a simple, short-cycled mechanism
  - Write a specification, in code and in the form of a unit test. The test verifies a functional unit of your code.
  - Demonstrate test failure.
  - Write code to meet the specification.
  - Demonstrate test success.
  - Refactor the code, to ensure that the system still has an optimally clean code base.
- Run all tests against the entire system at all time.

【reference】ppt09 p28; ppt10 p04

ii) A Student class should have a checkRegistration() method to return the registration status of the student. This method should take the String studentID as input parameter and return true if the student is registered. The student id "2015123456" is known as registered and the student id "2015654321" is known as unregistered. Write the test code using assertTrue and assertFalse method for these two test cases. Figure 1 show the API of the assertTrue and assertFalse method.

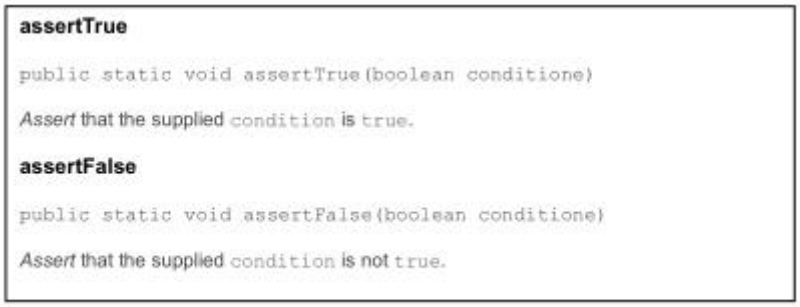


Figure 1

(4 Marks)

【Answer】

```
import org.junit.jupiter.api.Assertions.*;
   class StudentTest{
       void TestcheckRegistration(){
           assertTrue(checkRegistration("2015123456"));
           assertFalse(checkRegistration("2015654321"));
       }
   }
```

【考点】Test Class 书写

【解析】

本题为测试类书写应用问题。根据题意，我们需要书写一个测试方法，测试checkRegistration()。我们通过读题可知，这个方法输入是 studentID 字符串，然后返回 true（boolean 类型）。2015123456 注册过了，应该会返回 true；2015654321 没注册过，应该会返回 false。

我们需要使用 assertTrue 和 assertFalse 方法书写测试类。

assertTrue 和 assertFalse 输入参数要求都是 boolean 类型，而 checkRegistration 方法返回值也是 boolean,所以直接 checkRegistration(参数）即可作为条件填入，assert 后面输入参数。

按 Junit 的要求,开头 import 的内容要写,之后方法名 TestcheckRegistration,类型为 void。

据此分析答案为:

```
import org.junit.jupiter.api.Assertions.*;
    class StudentTest{
        void TestcheckRegistration(){
            assertTrue(checkRegistration(“2015123456”));
            assertFalse(checkRegistration(“2015654321”));
        }
    }
```

(4 Marks, import 1 mark, void TestcheckRegistration 1 mark, 剩下两句 1 for each)

iii) Write the method declaration of the checkRegistration() method correctly to pass the tests in ii). (2 Marks)

【Answer】

```
public boolean checkRegistration(String studentID){}
```

【考点】Test Class 书写

【解析】

Method declaration 就是方法函数开头函数名加上输入参数和大括号，大括号里的是方法体（不属于 declaration）

本题要求书写 checkRegistration 方法本身，根据测试类反推。我们可知，返回 true 就是返回类型 boolean，要输入 studentID 那就是 String studentID。

Method declaration 是:

```
public boolean checkRegistration(String studentID){}
```

(2 Marks, return type 1 mark, input parameter 1 mark)

如果写完整方法，那就是如下：

```
public boolean checkRegistration(String studentID){
    if(studentID.equals(“2015123456”){
        return true;
    }
    else if(studentID.equals(“2015654321”){
        return false;
    }
    return false;
} (2 Marks)
```

Question 2: (33 Marks)

a) A software product is in the final validation testing phase before being released to the general market. This software product has 9 functions that are accessible through 3 main menus. The validation testing is to verify whether the software product has met all requirements that were captured during the requirements gathering phase. Answer the following questions: (12 Marks)

i) What is the type of this software product and why? (2 Marks)

【Answer】

Application software. Because this software product perform a specific task.

【考点】Software Type

【解析】

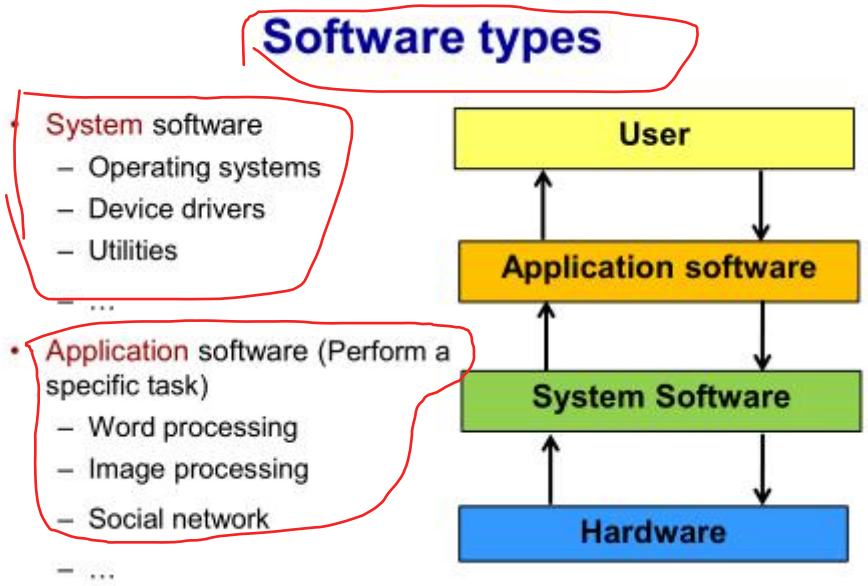
本题考查 software type，判断并解释理由。

回顾课件，我们可知 software type 分为两种，system software 和 application software。System software 偏向于底层系统应用，application software 偏向于纯应用。本题目的系统偏向纯应用而且有多界面多功能，会有具体的实现。所以属于 Application software.

理由很简单 Perform a specific task.

故答案为：

Application software. Because this software product perform a specific task. (2 Marks, 1+1. 答错类型直接 0 分)



【拓展】如果属于 System Software，必须和系统硬件相关，操作系统，设备驱动，组件之类的。否则就是 Application Software。

【reference】ppt01 p14

ii) Who should perform the tests and what testing technique should be used at this phase? Explain. (5 Marks)

【Answer】

At the component level, by the developers who developed the component. At the integration level, by independent testing engineer/team who were not involved with the construction of the system but have a detailed understanding of the whole system.

At this stage, the testing technique we use is Black-box testing because black-box testing focus on the functional requirements of the software, to test software requirements. Detailed saying, Scenario-based testing, which is a validation testing technique where you consider each requirement and derive a set of tests for that requirement.

【**考点**】Who tests the software, Testing Techniques 综合考察

【**解析**】

本题问谁测试这个软件，以及在当前阶段运用什么测试方式。

首先，谁测试这个软件？两个角度，component 和 integration level

At the component level, by the developers who developed the component.

At the integration level, by independent testing engineer/team who were not involved with the construction of the system but have a detailed understanding of the whole system. (2 Marks)

根据题目，我们知道该软件已经处于发布前最后阶段，而且测试的重心是是否满足所有的需求。Testing techniques 有两种，black-box 和 white-box

在最后阶段，如果要测试是否满足需求，肯定是 black-box testing。

At this stage, the testing technique we use is **Black-box testing** because black-box testing **focus on the functional requirements of the software**, to test software requirements. Detailed saying, **Scenario-based testing, which is a validation testing technique** where you consider each requirement and derive a set of tests for that requirement.

(5 Marks)

Who tests the software?

- At the component level
  - By the developers who developed the component.
  - Driven by delivery.
- At the integration level
  - By independent testing engineer/team who
    - were not involved with the construction of the system;
    - but have a detailed understanding of the whole system.
  - Attempt to break it.
  - Driven by quality.

Black/White-box Testing

In general, software is tested from two different perspectives:

- **Black-box testing** → to test software requirements
  - Also called behavioural testing.
  - Focus is on the functional requirements of the software.
- **White-box testing** → to test the internal program logic
  - Also called *glass-box testing* or *clear-box testing*.
  - Component level test case design.
  - Internal operations are performed according to specifications, and all internal components have been adequately exercised.

Scenario-based testing

- A typical black-box testing technique
- Requirements should be testable:
  - A test can be designed, observer can check that the requirement has been satisfied.
- A validation testing technique where you consider each requirement and derive a set of tests for that requirement.
- Used in system testing.

【reference】ppt09 p4 p21 p26

iii) What will a successful validation testing show? (2 Marks)

【**Answer**】A successful validation testing shows that the system operate as intended.

【**考点**】Testing Goals

【**解析**】

本题直白提问，一个成功的验证测试表明了什么？很简单，系统运行符合预期。A successful validation testing shows that the system operate as intended. (2 Marks)

【**注意区别**】如果问 Validation testing 的目的，没有提 successful，我们需要回答：To demonstrate that the software meets its requirements to both developer and customer.

【**总结**】Testing Goals

Validation Testing: 2 个重点

To demonstrate that the software meets its requirement to both customer and developer. 是否对于顾客和开发者符合预期

A successful validation testing shows that the system operate as intended. 成功的验证测试说明系统运行符合预期。

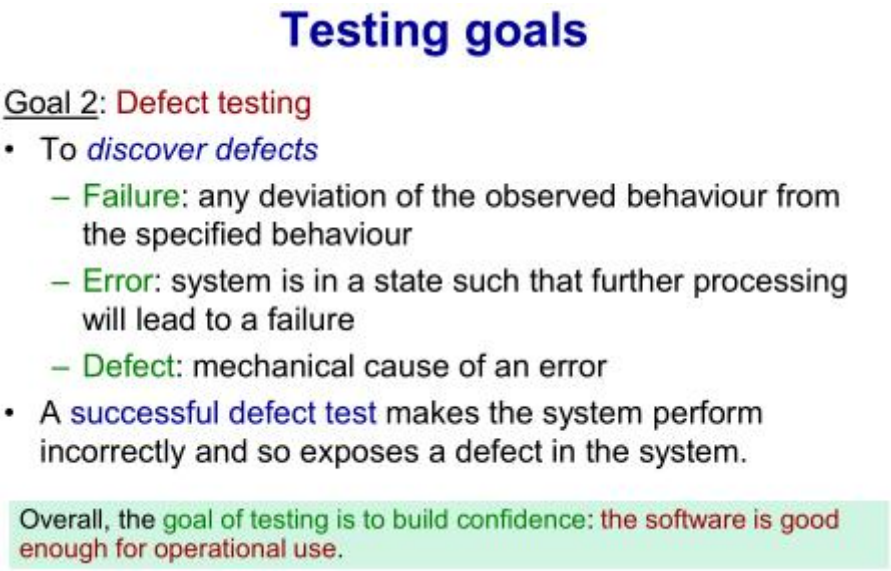
Defect Testing:

To discover defect. 发现错误

A successful defect test makes the system perform incorrectly and so exposes a defect in the system. 成功的查错测试让系统的错误暴露。

Overall, the goal of testing is to build confidence: the software is good enough for operational use. 测试为了建立信心，软件能用





【*reference*】*ppt09 p5 p6*

iv) Design a testing policy for this validation testing. (3 Marks)

【Answer】

1) All functions accessed through menus should be tested.

2) Combinations of functions accessed through the same menu should be tested.

3) When user input is required, all functions must be tested with correct and incorrect input.

【考点】Testing Policy

【解析】

本题考察 Testing policy 的设计。

要设计 Testing Policy，我们需要知道 testing policy 是什么。是选择系统测试的方式，什么要测，要怎么测这方面的内容。开放问题。

The validation testing is to verify whether the software product has met all requirements that were captured during the requirements gathering phase.

回归原题，这个 validation testing 是要去检验是否满足所有当时捕获的需求。但软件产品没有说具体什么是和什么相关，有 3 个 menu 和 9 个 function，那么 menu 和 function 必须都要测试。此外，用户输入，正确与不正确的都要测试到。

那么，答案为：

4) All functions accessed through menus should be tested.

所有经过 menu 的功能必须测试到。

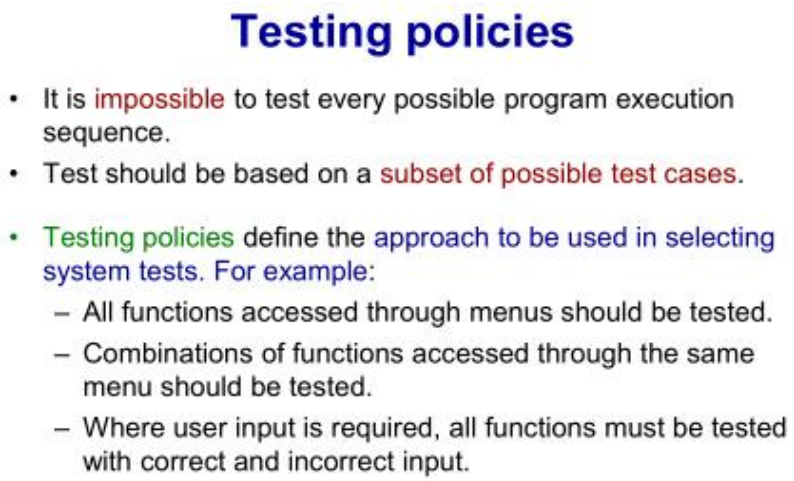
5) Combinations of functions accessed through the same menu should be tested.

组合的功能经过同一个界面必须被测试到。

6) When user input is required, all functions must be tested with correct and incorrect input.

所有需要用户输入的功能，正确与不正确的都需要测试到。

(3 Marks, 1 for each)



【reference】ppt09 p7

b) An online exam system was released for operational use and was used for the first time to conduct an automated online exam for a university. There was no preparation for an alternative paper-based exam in case the automated online exam fails. After the online exam was conducted, there were errors in the questions and some students complained that the user interface was not good and the system was not easy to use. Answer the following questions: (11 Marks)

i) Using appropriate software engineering management strategies and terms, discuss how the problems encountered above could have been professionally identified, managed and avoided. (6 Marks)

【Answer】

We use Risk Management to identify, manage and avoid the problem.

We should do as follows:

Identify project, product and business risks.

Assess the likelihood and consequences of these risks.

Draw up plans to avoid or minimise the effect of the risk.

Monitor the risk throughout the project.

For avoiding the problem, we should be careful about using new or unfamiliar products, balance any benefit obtained from risk they will not perform as expected. Actively monitor performance, do not hide problems.

【考点】Project Management

【解析】

本题的情景为网络线上考试，相信大家都是熟悉且经历过的。

从文中我们可以提取出如下问题：

1、第一次在大学考试使用自动化线上考试，

没有预案——在系统出问题的时候，没有准备备用纸质考试的方案；

2、考试题目出现问题，一些学生抱怨界面不好，系统不好用；

该问题要求我们使用合适的软件工程管理策略，对于上述问题如何专业地识别，管理和避免。

不难根据问题中提出对用户界面的不友好问题以及没有在错误发生时有预案，可判断，这是一个风险管理策略需要应对的。具体根据风险管理策略的理论结合实际说明即可。

We use Risk Management to identify, manage and avoid the problem.

We should do as follows:

Identify project, product and business risks.

Assess the likelihood and consequences of these risks.

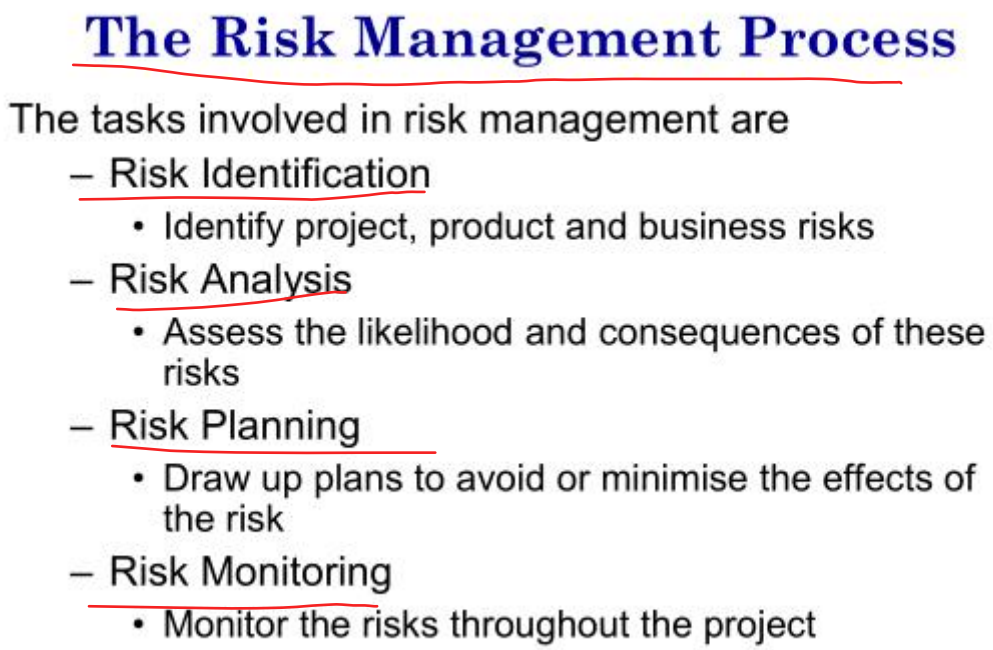
Draw up plans to avoid or minimise the effect of the risk.

Monitor the risk throughout the project.

For avoiding the problem, we should be careful about using new or unfamiliar products, balance any benefit obtained from risk they will not perform as expected. Actively monitor performance, do not hide problems. (6 Marks)

核心要点：

识别项目产品商业风险；评估可能性和风险结果；制定计划规避风险；监控风险。如何回避，谨慎使用不熟悉的产品并积极主动监控不隐藏问题。



Avoidance Strategies

- Be careful about using new or unfamiliar products (tools, software, hardware), balance any benefit obtained from risk they will not perform as expected
- But keep awareness of new products, do not stick to old tools and systems out of habit or unwillingness to change
- Ensure staff are well trained, avoid having just one person who has skills and/or knowledge of a vital topic
- Ensure working environment is pleasant, so staff are not tempted to move
- Maintain awareness of the market, new technologies, competing products
- Ensure senior management are well informed on progress of project and its importance to the organisation's goals
- Actively monitor performance, be prompt if there is any slippage, do not "hide" problems from senior management or staff

【reference】ppt14 p9 p10

ii) Describe how to ensure that the online exam system is "good enough" so that it can be used in future exams. (5 Marks)

【Answer】

To ensure that the online exam system is "good enough", we should do as below:

1、Achieved between meeting students' demand of user interface and system operations.

2、Giving guarantees about performance of the system.

3、We cannot guarantee that the system is completely free from error. When the system have error, alternative plan should be drawn to face the problem.

【考点】"Good Enough" Software

【解析】

本题问如何去保证线上考试系统是足够好，能够在未来使用。

我们需要知道什么时候足够好,根据足够好的原则结合线上考试文字中提及的内容，进行构建答案。

To ensure that the online exam system is “good enough”, we should do as below:

1、Achieved between meeting students’ demand of user interface and system operations.

2、Giving guarantees about performance of the system.

3、We cannot guarantee that the system is completely free from error. When the system have error, alternative plan should be drawn to face the problem.

(5 Marks)



【reference】ppt14 p18

c) Answer the following questions about open source software. (10 Marks)

i) Define open source and open source software (OSS). (4 marks)

【Answer】

Open source means that source code being publicly available.

Open source software (OSS) means that software which is free of charge, and free of legal restrictions on how it can be used.

【考点】Open Source

【解析】

本题要求界定 open source 和 open source software，直白问定义。直接写定义即可。

Open source means that **source code** being **publicly available**. 代码开源

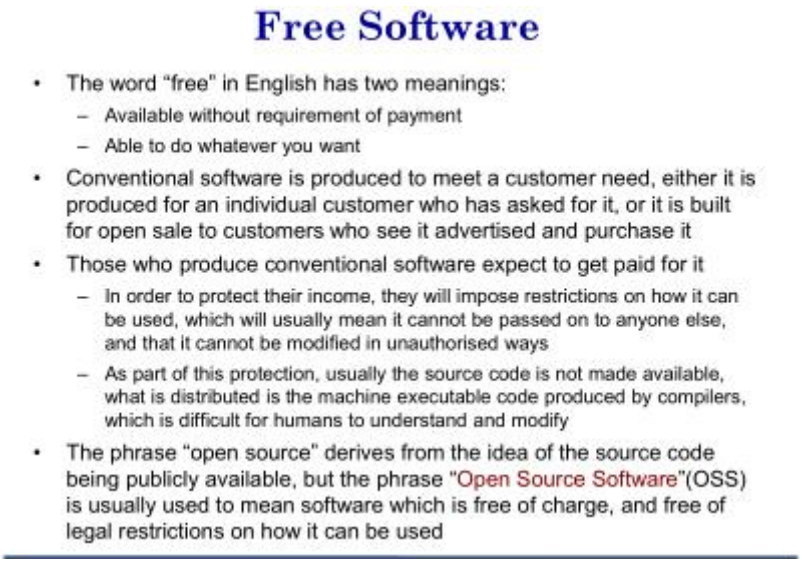
Open source software (OSS) means that software which is **free of charge**, and **free of legal restrictions on how it can be used**.

免费+免于使用限制

(4 Marks, 2 for each)

区别：前者针对源代码，后者针对软件。

记忆方式：OS 源代码公开可见（开源）；OSS 软件，免费，免怎么使用的限制。2 个 free。



Free Software

- The word "free" in English has two meanings:
  - Available without requirement of payment
  - Able to do whatever you want
- Conventional software is produced to meet a customer need, either it is produced for an individual customer who has asked for it, or it is built for open sale to customers who see it advertised and purchase it
- Those who produce conventional software expect to get paid for it
  - In order to protect their income, they will impose restrictions on how it can be used, which will usually mean it cannot be passed on to anyone else, and that it cannot be modified in unauthorised ways
  - As part of this protection, usually the source code is not made available, what is distributed is the machine executable code produced by compilers, which is difficult for humans to understand and modify
- The phrase "open source" derives from the idea of the source code being publicly available, but the phrase "Open Source Software"(OSS) is usually used to mean software which is free of charge, and free of legal restrictions on how it can be used

【reference】ppt17 p2

ii) The analyst firm "Vision Mobile" has developed a way of measuring the openness of OSS projects, and claimed "Android" showed the least openness of the major OSS projects they investigated. State three reasons for "Vision Mobile" giving this result. (6 Marks)

【Answer】

The key aspect of their claim is their suggestion that an open governance model is as important to a project being truly open as the licensing terms. In the investigation report, we know that Android has the lowest Open Governance Index among those OSS projects like Symbian, Qt, Mozilla, etc. Android is closely controlled by Google, which gives priority access to specific developer groups and organizations, and does not make public its internal decision-making processes.

Google does not provide public information regarding meetings held and decisions made on the development of Android.

Google owns the Android trademark, and will not permit it to be used for devices which use Android code but do not pass a set of tests, the Compatibility Test Suite(CTS) which it controls.

【考点】Open Source

【解析】

本题要求列举 Vision Mobile 给 android 这个理由的原因。

首先，我们可知 open governance model 是最重要的评判标准。

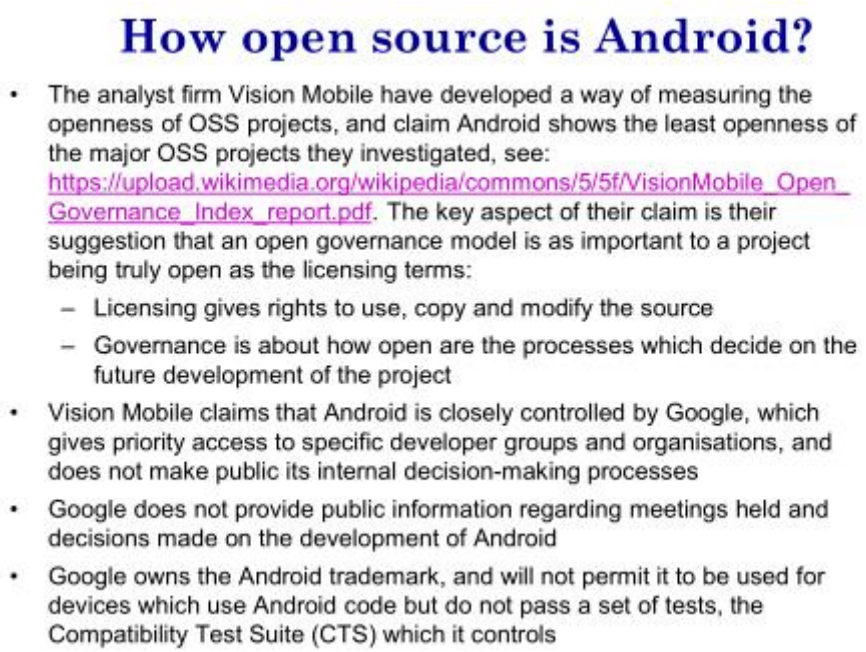
相比其他的 OSS projects，Android 位居最后一名。

The key aspect of their claim is their suggestion that an open governance model is as important to a project being truly open as the licensing terms. In the investigation report, we know that Android has the lowest Open Governance Index among those OSS projects like Symbian, Qt, Mozilla, etc. Android is closely controlled by Google, which gives priority access to specific developer groups and organizations, and does not make public its internal decision-making processes.

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(6 Marks)



【reference】ppt17 p22

https://upload.wikimedia.org/wikipedia/commons/5/5f/VisionMobile_Open_Governance_Index_report.pdf

Question 3: (34 Marks)

a)

Suppose we have a Java class OurGame which has a constructor with this signature:

```
public OurGame(Information inf)
```

And several methods including one with this signature:

```
public boolean move(Character c, Position p)
```

Also a class Account that has a method with the following signature:

```
public boolean withdraw(int amount)
```

We want a modified version of OurGame called PayedGame that has a variable of type int called cost and a variable of type Account called acc. The values of these variables are set by the constructor of PayedGame. It

represents a form of game where making a move has to be paid for, so the method move needs to be changed to implement that.

Paying for the move is given by the method call acc.withdraw(cost). If that call returns false, then no move can be made, so the call to the method move should return false without doing anything else. Otherwise the call move should work the same as the call move in the class OurGame, and every other method in PayedGame should work like the same method in OurGame.

The class PayedGame could be implemented just by copying the code from OurGame, changing the name of the class to PayedGame, and changing the code inside it to provide the extra features required. Explain why this is not a good way of providing the PayedGame class, particularly if it is part of a large project which already makes use of the OurGame class. (6 Marks)

【Answer】

Providing the PayedGame class which implemented just by copying the code from OurGame violates the don't repeat yourself (DRY) design principle. For a large project, especially when the method OurGame has many methods with large size of code. After duplicate the code to PayedGame, it'll have a significant impact on the operation of the software, causing the efficiency lower and cost a lot more memory to store and run the code. Besides, suppose finding a mistake in the code, you will have to go through all the places where you used "cut and paste" to modify the code, consistency is hard to guarantee with manual operation, also, working load will be a lot bigger.

Code will be not representative for programmer to read and programmer is harder to do the maintenance of the code.

【考点】Design Principles

【解析】

本题非常长的文字是介绍这个软件的情况，主要聚焦于问题。

问题是为什么 PayedGame 直接复制 OurGame 的代码，改个类名，加上需要的额外特点不好。尤其是在已经大项目已经使用 OurGame 类的情况下。

后半句，很明显可以认为 PayedGame 复制代码已经是重复自己的行为，违背原则 Don’t repeat yourself. (DRY)。这是核心需要点明的。

此外，结合该原则的相关内容，进行展开解释。

Providing the PayedGame class which implemented just by copying the code from OurGame violates the don’t repeat yourself (DRY) design principle. For a large project, especially when the method OurGame has many methods with large size of code. After duplicate the code to PayedGame, it’ll have a significant impact on the operation of the software, causing the efficiency lower and cost a lot more memory to store and run the code. Besides, suppose finding a mistake in the code, you will have to go through all the places where you used “cut and paste” to modify the code, consistency is hard to guarantee with manual operation, also, working load will be a lot bigger.

Code will be not representative for programmer to read and programmer is harder to do the maintenance of the code. (6 Marks, DRY 1 mark, 2+2+1).

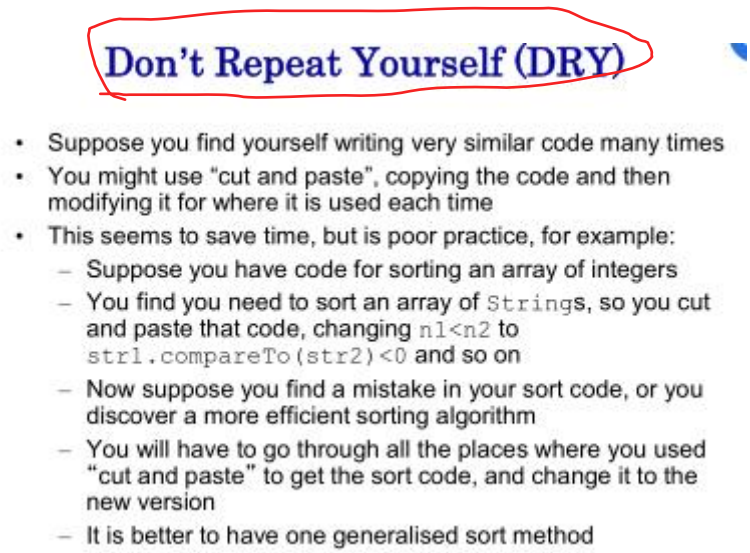
基本逻辑，违反 DRY 原则。

主要从三方面说，

1 重复代码影响运行效率；

2 如果出错修改可能造成一致性问题而且修改的工作量增大很多；

3 代码也会更难被程序员理解和维护。



【reference】ppt15 p57

b) Explain how inheritance could be used to provide class PayedGame. (10 Marks)

【Answer】

1、We declare the class PayedGame as a subclass of OurGame, which is superclass of PayedGame.

2、We only add extra features (methods and instance variables that they use) in the subclass PayedGame, attribute cost and account.

3、We change the method move, which is worked as a different way from the superclass OurGame. Overriding the method move.

By doing so, we writing a subclass PayedGame by modifying the way the code of the superclass OurGame works without changing that actual code.

【考点】Inheritance

【解析】

本题问我们如何利用继承提供 PayedGame 类。

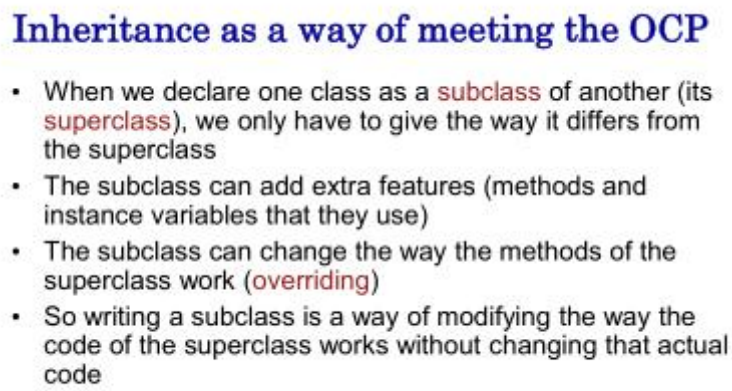
我们已经在前序知道 PayedGame 类违反了 DRY，而 DRY 是一种特殊的 OCP 原则，那么如何利用继承去满足 OCP 原则的内容结合具体该题的情况是我们需要作答的内容。

1、We declare the class PayedGame as a subclass of OurGame, which is superclass of PayedGame.

2、We only add extra features (methods and instance variables that they use) in the subclass PayedGame, attribute cost and account.

3、We change the method move, which is worked as a different way from the superclass OurGame. Overriding the method move.

By doing so, we writing a subclass PayedGame by modifying the way the code of the superclass OurGame works without changing that actual code. (10 Marks)



【reference】ppt15 p59

c) Write the code that implements the class PayedGame in b) using inheritance. (10 Marks)

【Answer】

```
public class PayedGame extends OurGame(){
    private int cost;
    private Account acc;
    public PayedGame(int cost, Account acc){
        this.cost=cost;
        this.acc=acc;
    }
    public boolean move(Character c, Position p){
        if(acc.withdraw(cost)){
```

```
            super.move(c,p);
            return true;
        }
        else{
            return false;
        }
    }
}
```

【考点】Inheritance 代码实现

【解析】

解决 Repeat 代码的其中一种方式是使用 Inheritance，即使用继承方式实现 PayedGame 类。

根据题意和材料介绍，转化成对应的代码内容。拆分句子进行转化代码。

1、We want a modified version of OurGame called PayedGame that has a variable of type int called cost and a variable of type Account called acc.

Attribute 需要两个， private int cost; private Account acc; 通常类内访问加 private。

2、The values of these variables are set by the constructor of PayedGame.

所以 constructor 方法：

```
public PayedGame(int cost, Account acc){
    this.cost=cost;
    this.acc=acc;
}
```

3、It represents a form of game where making a move has to be paid for, so the method move needs to be changed to implement that.

提示我们要修改 move，所以 move 方法被子类 override，需要重写 move 方法。

4、Paying for the move is given by the method call acc.withdraw(cost). If that call returns false, then no move can be made, so the call to the

method move should return false without doing anything else. Otherwise the call move should work the same as the call move in the class OurGame.

要求根据acc.withdraw(cost)的情况来判别move的情况。

如果返回false，就没有移动操作不做任何事情，没有代码；如果返回true，和OurGame实现一样的move方法。注意输入参数,c和p需要,因为调用父类OurGame中需要这两个参数，对于cost，该类中已有cost的属性，在这里可直接用。

```
public boolean move(Character c, Position p){
    if(acc.withdraw(cost)){
        super.move(c,p);
        return true;
    }
    else{
        return false;
    }
}
```

5、And every other method in PayedGame should work like the same method in OurGame.

继承OurGame即可体现其他的方法，子类不需要重新写。

那么由此，我们可以得到答案：

```
public class PayedGame extends OurGame(){
    private int cost;
    private Account acc;
    public PayedGame(int cost, Account acc){
        this.cost=cost;
        this.acc=acc;
    }
    public boolean move(Character c, Position p){
        if(acc.withdraw(cost)){
            super.move(c,p);
```

```
            return true;
        }
        else{
            return false;
        }
    }
} (10 Marks)
```

d) An alternative way of implementing PayedGame would be to use the Decorator design pattern. Explain how this could be done. Explain how it differs in the way it can be used from the implementation of PayedGame using inheritance.

Your explanation should be on the basis that there is an interface type called Game which has all the method signatures of class OurGame, and class OurGame is declared as implements Game. (8 Marks)

【Answer】

How this could be done:

1、There is Game interface type existed, OurGame implements this interface, which involves in the Decorator design pattern.

2、In the PayedGame class, variable is set in the constructor.

3、For the PayedGame method, has plus some extra work than the OurGame class, which is implemented in the interface Game.

Differ from using inheritance:

Inheritance do not need to implement an interface, only the subclass and superclass, subclass is PayedGame, superclass is OurGame. We don't need the Game interface by using inheritance and can achieved the same effect as decorate design pattern.

【考点】Decorate Design Pattern

【解析】

本题考查对 Decorate Design Pattern 的深度理解。解释利用这种设计模式如何实现 PayedGame 方法，以及这种模式实现和 inheritance 实现的不同点。

同时要求基于存在界面类 Game 的前提下进行解释。

How this could be done:

1、There is Game interface type existed, OurGame implements this interface, which involves in the Decorator design pattern.

2、In the PayedGame class, variable is set in the constructor.

3、For the PayedGame method, has plus some extra work than the OurGame class, which is implemented in the interface Game.

Differ from using inheritance:

Inheritance do not need to implement an interface, only the subclass and superclass, subclass is PayedGame, superclass is OurGame. We don’t need the Game interface by using inheritance and can achieved the same effect as decorate design pattern.

(8 Marks, 2 for each, theory1+specification explanation1)